

The Macroeconomic Consequences of Exchange Rate Depreciations

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BGGMP Reading Group

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No agreement on how exchange rate depreciations affect the economy

- Expansionary (expenditure switching)
- Contractionary (real income effect, balance sheet effect)
- Disconnected from macro fundamentals

What is driving exchange rate depreciations?

- Exchange rate is endogenous and the answer may depend on the underlying shock

This paper

New identification strategy for exchange rate shock

- Compare countries that peg their currency to the US dollar versus let them float

Find that depreciations are expansionary

- Not led by expenditure-switching or looser monetary policy

Financially driven exchange rate model

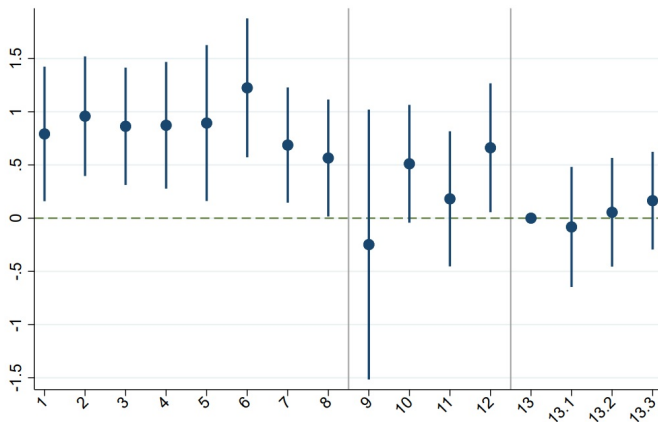
- Key assumptions: limits to arbitrage, households and firms have access to international financial market
- Two financial shocks: UIP shock and capital flight shock
- Matches the response to an UIP shock and unconditional business cycle moments

Table of Contents

- 1 Empirical Analysis
- 2 The Model
- 3 Quantitative Model: Expansionary Depreciation
- 4 Quantitative Model: Exchange Rate Disconnect

- Idea: compare peg in float currencies responses to depreciation of US dollar
 - Identifies an exchange rate change not driven by domestic conditions
- Use US dollar nominal effective exchange rate
 - Trade-weighted exchange rate relative to 24 countries, 1973-2019, BIS
 - Those countries excluded from the sample of peggers and floaters
- Exchange rate regimes
 - Classification by Ilzetzi, Reinhart and Rogoff (2019)
 - Check how countries from different categories comove with the US dollar

Comovement with US Dollar by Exchange Rate Regime



- 1 - peg
- 12 - managed floats
- 13 - free floating
- 13.1-3 - anchor other than US dollar

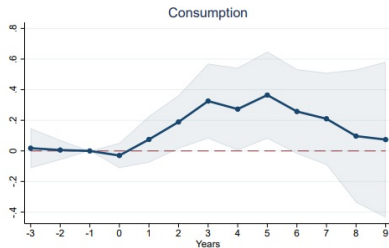
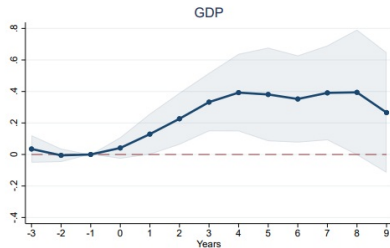
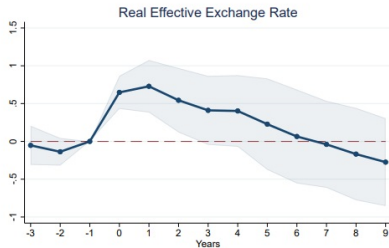
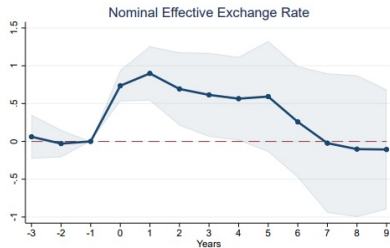
- Local projection

$$y_{i,t+h} - y_{i,t-1} = \alpha_{i,h} + \alpha_{r(i),t,h} + \beta_h \text{Peg}_{i,t} \times \Delta e_{USD,t} + \Gamma'_h \mathbf{x}_{i,t-1} + \gamma_h \text{Peg}_{i,t} + \epsilon_{i,t,h}$$

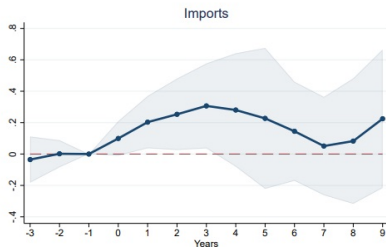
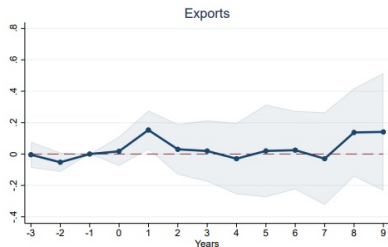
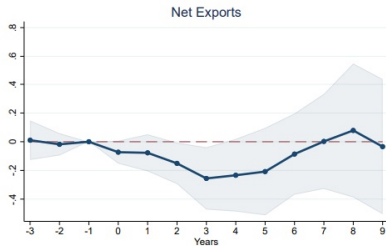
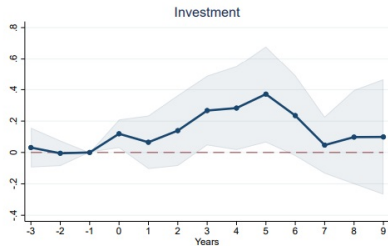
- Identifying assumption: pegs and floats are identically exposed to the shocks correlated with the US nominal effective exchange rate
 - From optimal currency area theory, pegs are exposed more
 - If so, the results will be biased towards contractionary depreciations

More

Pegs expand relative to floats after dollar depreciation



...but net export falls



...and monetary policy is contractionary

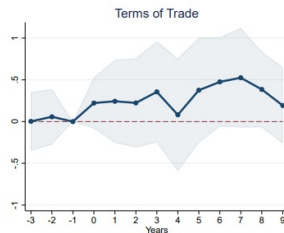
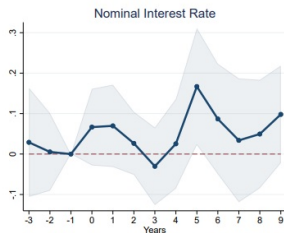


Table of Contents

- 1 Empirical Analysis
- 2 The Model
- 3 Quantitative Model: Expansionary Depreciation
- 4 Quantitative Model: Exchange Rate Disconnect

Overview of the Model

- Open economy New Keynesian model with financial shocks and frictions
- Three countries: the US, pegger, floater
- Builds on Itshoki Mukhin 2021
 - Exchange rate is driven by a financial shock to the demand of noise traders
 - UIP deviations due to limits to arbitrage
 - Exchange rate is disconnected from macro fundamentals
- Key differences:
 - Firms and households have access to international financial markets
 - Two different financial shocks (UIP and capital flight shock)

- Household consume the final good and provide labor elastically

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{(C_{it} - hC_{it-1})^{1-\sigma}}{1-\sigma} - \frac{n_{it}^{1+\nu}}{1+\nu} \right]$$

- Consumption has home bias

$$C_{it} = \left((1-\alpha)^{1/\eta} (c_{iit})^{\frac{\eta-1}{\eta}} + \alpha^{1/\eta} \int_0^1 (c_{jit})^{\frac{\eta-1}{\eta}} dj \right)^{\frac{\eta}{\eta-1}}$$

- The budget constraint is

$$C_{it} + a_{it} = a_{it-1}(1 + r^h) + (1 - \tau_i^n) W_{it} N_{it} / P_{it} + T_{it}$$

where a_{it} is the value of the portfolio of assets

- A firm produces a homogeneous country-specific good

$$Y_{it} = A_{it} (K_{it}^{\varkappa} N_{it}^{1-\varkappa})^{1-\omega} X_{it}^{\omega}$$

- N_{it} is the labor demand
- X_{it} is intermediate inputs (the same CES basket as final consumption good)
- Perfectly competitive in home market, so set $p_{it}^{mc} = mc_{it}$ to sell product to the price-setting firm

Production firms

- The earnings of a firm are

$$D_{it} = \frac{1}{P_{it}} \left[p_{it}^{mc} Y_{it} - P_{it} l_{it} \left(1 + S \left(\frac{l_{it}}{l_{it-1}} \right) \right) - W_{it} N_{it} - P_{it} X_{it} + \Pi_{it}^p \right]$$

where Π_{it} is profit from owning price-setting firms

- The firm maximizes its value

$$V_{it} = D_{it} + \mathbb{E}_t \sum_{s=t}^{\infty} \frac{D_{is+1}}{(\Pi_{k=0}^t (1 + r_{ik+1}^f))}$$

where $(1 + r_{ik+1}^f)$ is financing cost

- Aggregate productivity follows AR(1) process

- The return on domestic equity is r_{it+1}
- The return on foreign currency bonds is

$$(1 + r_{ijt+1}) = (1 + r_{jt+1}) \frac{Q_{jit+1}}{Q_{jit}}$$

- Households and firms choose portfolio shares incurring adjustment costs when deviating from the steady-state portfolio
 - Steady-state portfolio is exogenous
 - Assume that country net foreign position is zero in steady state
 - Steady-state shares are equal for households and firms
- Due to financial frictions, $r_{it+1} \neq r_{ijt+1}$

Households maximize (analogous for firms)

$$\max_{\{s_{ijt}^h\}} \mathbb{E}_t \left[\left(1 - \int_0^1 s_{ijt}^h dj \right) (1 + r_{it+1}) + \int_0^1 \left(s_{ijt}^h (1 + r_{ijt+1}) - \Phi_{ij}^h(s_{ijt}^h) \right) dj \right]$$

- s_{ijt}^h is portfolio share in country j bonds
- adjustment cost: $\Phi_{ij}^h(s_{ijt}) = \frac{\Gamma^h}{2\bar{s}_{ij}} (s_{ijt}^h - \bar{s}_{ij})^2$
- $\bar{s}_{ij}$ is steady state portfolio share

- Noise traders with positions ψ_{jt} create UIP shock

$$\psi_{jt} = \rho^{\psi} \psi_{jt-1} + \epsilon_{jt}^{\psi}$$

- International bonds arbitrageurs specialize in carry trade between currency j and US dollar
 - Risk averse, so arbitrage capacity is limited

International bonds arbitrageurs solve

$$\max_{B'_{Ujt}} -\mathbb{E}_t \frac{1}{\gamma} \exp \left(-\gamma \left[\frac{\tilde{R}_{Ujt+1}}{P_{U,t+1}} B'_{Ujt} \right] \right)$$

where

- B'_{Uji} is the carry trade position (borrow in U , invest in j)
- The nominal return per dollar is $\tilde{R}_{Ujt+1} \equiv (1 + i_{jt}) \frac{\mathcal{E}_{jU,t+1}}{\mathcal{E}_{jU,t}} - (1 + i_{Ut})$

More

Deviations from UIP

- For peggers, UIP holds

$$(1 + i_{jt}) = \mathbb{E}_t (1 + i_{Ut}) \frac{\mathcal{E}_{jUt+1}}{\mathcal{E}_{jUt}} \quad \text{for } j \in P$$

- No exchange rate risk means perfect arbitrage even with risk-averse arbitrageurs
- For floaters, there is a deviation $\Omega(\{NFA_{kt}\}_k, \psi_{jt})$ from UIP [More](#)

$$(1 + i_{j,t}) = \mathbb{E}_t (1 + i_{U,t}) \frac{\mathcal{E}_{jU,t+1}}{\mathcal{E}_{jU,t}} \exp(\Omega(\{NFA_{kt}\}_k, \psi_{jt}))$$

- Wages are reset by labor union with probability $1 - \delta_w$
- Prices are reset by price-setting with probability $1 - \delta_p$
 - Set prices in domestic currency for domestic market
 - When selling abroad, sets prices in producer currency

- Central banks in US and F follow an interest rate rule:

$$\ln(1 + i_{jt}) = \ln \bar{R} + \rho^m \ln(1 + i_{jt-1}) + (1 - \rho^m) \phi_\pi \pi_{jt} + \epsilon_{jt}^m$$

for $j \in \{F, U\}$

- Central bank in P fix nominal exchange rate to US dollar:

$$\mathcal{E}_{jUt} = \bar{\mathcal{E}}_{jU}$$

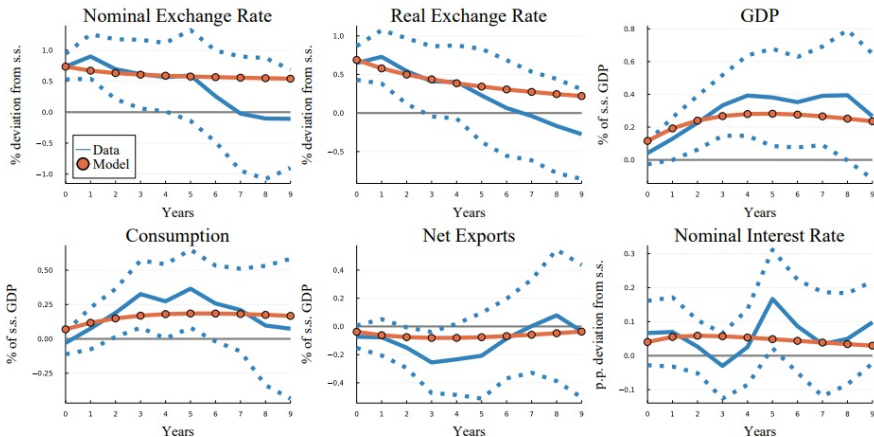
for $j \in P$

Table of Contents

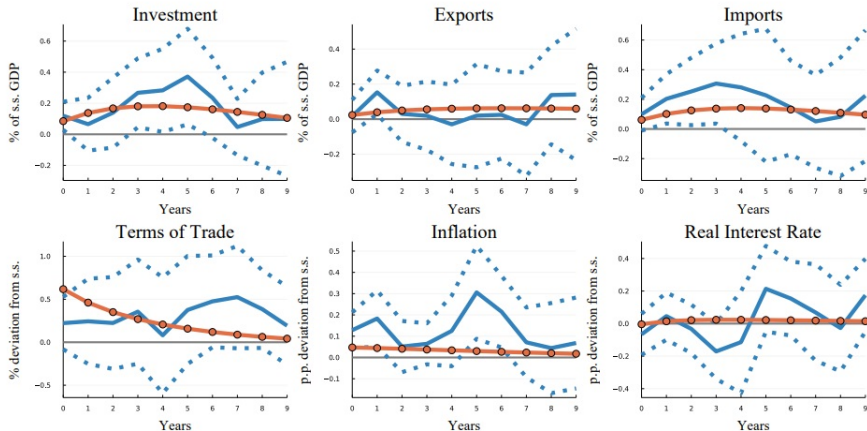
- 1 Empirical Analysis
- 2 The Model
- 3 Quantitative Model: Expansionary Depreciation
- 4 Quantitative Model: Exchange Rate Disconnect

- Regions size: $|U| = 0.3$, $|F| = 0.5$, $|P| = 0.2$
- Net foreign asset position for each country is zero
- $\bar{s} = 0.52$ targets the average value of gross foreign assets
- The composite parameter for the strength of financial frictions $\Gamma = 0.001$ is small following Itskhoki Mukhin (2021)
 - UIP deviations due to changes in NFA are small
 - UIP shocks can still have large effects if the variance sufficiently large
- Other parameters standard [More](#)

Pegs expand relative to floats after a US UIP shock

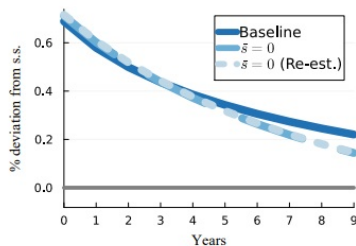


Foreign credit is the key channel

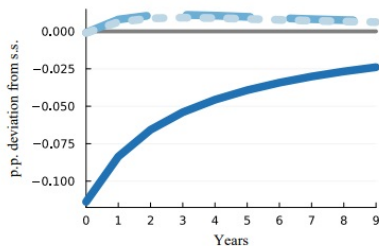


No expansion without foreign borrowing ($\bar{s} = 0$)

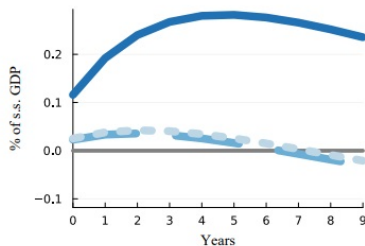
Real Exchange Rate



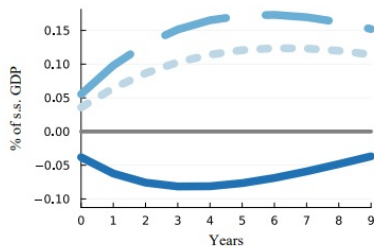
Household and Firm Financial Discounts



GDP



Net Exports



- As UIP shock is AR(1), depreciation today means that the agents expect appreciation next period
- So, foreign currency borrowing becomes cheaper
- The change in financing rate for firms is

$$d \ln(1 + r_{it+1}^f) = d \ln(1 + r_{it+1}) - \bar{s} d\Omega(NFA_{ktk}, \psi_{Ft})$$

- Firms increase investment
- Households increase consumption, which raises import

Table of Contents

- 1 Empirical Analysis
- 2 The Model
- 3 Quantitative Model: Expansionary Depreciation
- 4 Quantitative Model: Exchange Rate Disconnect

Exchange rate disconnect

- Do expansionary depreciations contradict the exchange disconnect facts?
 - Exchange rates are very volatile and barely correlated with other real variables
 - Relative consumption is negatively correlated with real exchange rate
- Not necessarily
 - Disconnect facts: unconditional moments
 - Expansionary depreciation: UIP shock
- Introduce the second financial shock - capital flight shock

Capital flight shock

- Introduce a bank intermediation wedge ζ_{it} between the return on bonds from country j for agents from countries i and j

$$(1 + r_{ijt+1}) = (1 + r_{jt+1}) \frac{Q_{jit+1}}{Q_{jit}} \exp(\zeta_{it})$$

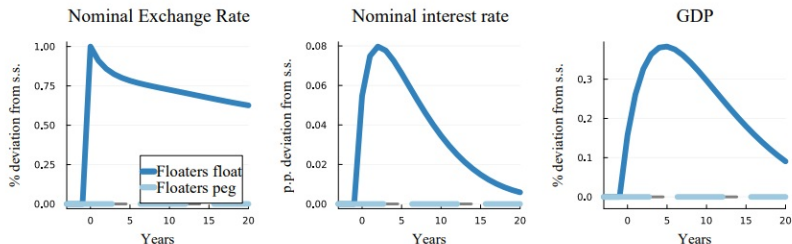
- The change in financing rate for firms is

$$d \ln(1 + r_{it+1}^f) = d \ln(1 + r_{it+1}) + \bar{s} \zeta_{it} - \bar{s} d\Omega(\{NFA_{kt}\}_k, \psi_{jt}, \{\zeta_{kt}\}_k)$$

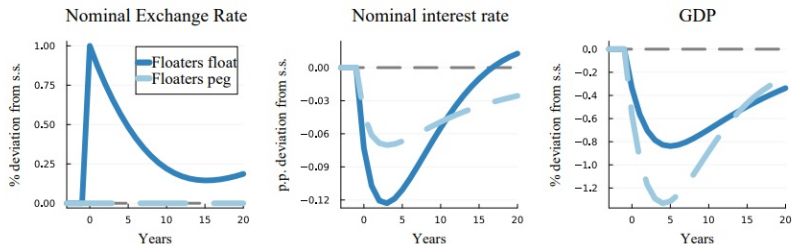
- Two effects
 - Directly increases borrowing costs
 - Capital outflow causes depreciation

Back

UIP vs capital flight shocks



(a) Impulse Response to a UIP Shock



Unconditional moments

	Data	Model						
		(1)	(2)	(3)	(4)	(5)	(6)	(7)
		(ψ, ζ)	(ψ, A)	ψ	ζ	A	m	(ψ, A)
		Baseline						$\bar{s} = 0$
A. Volatility								
std(ΔNER)	0.114	0.114	0.114	0.169	0.082	0.021	0.158	0.114
std(ΔRER)	0.091	0.111	0.110	0.163	0.082	0.021	0.157	0.112
std(ΔGDP)	0.037	0.037	0.037	0.037	0.037	0.037	0.037	0.037
std(ΔC)	0.042	0.039	0.022	0.029	0.042	0.014	0.017	0.016
std(ΔNX)	0.032	0.021	0.019	0.014	0.024	0.022	0.006	0.026
std($\Delta(1+i)$)	0.031	0.008	0.011	0.010	0.007	0.012	0.178	0.012
B. Correlation								
corr($\Delta RER, \Delta NER$)	0.712	0.992	0.987	0.989	0.996	0.868	1.000	0.991
corr($\Delta RER, \Delta GDP$)	-0.068	-0.068	0.490	0.603	-0.603	0.882	0.689	0.238
corr($\Delta RER, \Delta C$)	-0.137	-0.134	0.535	0.567	-0.544	0.597	0.676	-0.088
corr($\Delta RER, \Delta NX$)	0.213	0.126	-0.165	-0.589	0.481	0.955	-0.670	0.371
corr($\Delta RER, \Delta(1+i)$)	0.130	0.087	0.413	0.846	-0.784	-0.958	-1.000	0.092

Conclusion

- The paper introduces a novel identification strategy for exchange rate shock
- Shows that depreciations are expansionary
- Builds a model with two financial shocks reconciling this evidence with unconditional moments

Table of Contents

5 Appendix

Floats and pegs

Table 1: How Do Pegs Differ from Floats

Variable	No Controls	Time FE	Region X Time FE
Log Population	-0.02 (0.31)	-0.09 (0.31)	0.74* (0.39)
Log Real GDP Per Capita	0.36 (0.22)	0.32 (0.22)	-0.17 (0.23)
Export to GDP	-0.01 (0.04)	-0.01 (0.04)	0.00 (0.04)
Import to GDP	-0.03 (0.04)	-0.03 (0.04)	-0.03 (0.04)
Export Share to the US	0.04*** (0.01)	0.04*** (0.01)	-0.00 (0.01)
Import Share to the US	0.05*** (0.01)	0.05*** (0.01)	0.00 (0.00)
NFA to GDP	0.05 (0.18)	0.06 (0.19)	-0.10 (0.26)
Inflation Rate (p.p.)	-0.89 (1.51)	-0.65 (1.41)	2.21*** (0.69)
TBill Rate (p.p.)	1.01 (0.84)	0.89 (0.90)	2.86*** (0.96)
Commodity Exports to GDP	0.05* (0.03)	0.06** (0.03)	0.04 (0.03)
Commodity Imports to GDP	0.01 (0.02)	0.01 (0.02)	-0.01 (0.02)

Portfolio problem solutions

- Households:

$$s_{ijt}^h - \bar{s}_{ij} = \frac{\bar{s}_{ij}}{\Gamma^h} [\mathbb{E}_t (1 + r_{ijt+1}) - \mathbb{E}_t (1 + r_{it+1})]$$

- Firms

$$s_{ijt}^f - \bar{s}_{ij} = \frac{\bar{s}_{ij}}{\Gamma^f} [\mathbb{E}_t (1 + r_{ijt+1}) - \mathbb{E}_t (1 + r_{it+1})]$$

- Arbitrageurs

$$B_{Ujt}^I = \frac{1}{\Gamma^B} [\ln (1 + i_{jt}) - \ln (1 + i_{Ut}) + \mathbb{E}_t \Delta \ln \mathcal{E}_{jut+1}]$$

Back

- UIP deviation is

$$\Omega(\{NFA_{kt}\}_k, \psi_{jt}) \equiv -\Gamma \left[\left(1 - \int \bar{s}_{ji} di\right) NFA_{jt} + \int \bar{s}_{ij} NFA_{it} di + n^\psi \psi_{jt} \right]$$

where Γ is the composite parameter for financial frictions

$$\Gamma \equiv 1 / \left(\frac{1}{\Gamma^B} + \left[\frac{1}{\Gamma^h} + \frac{1}{\Gamma^f} \right] \frac{\bar{a}}{\bar{\beta}} \int_{i \in \{P, U\}} (\bar{s}_{ji} + \bar{s}_{ij}) di \right)$$

Back

Table B.3: Calibration of Parameters

Parameter	Description	Value	Note/Source
A. Externally Assigned Parameters			
β	Discount factor	0.96	Annual interest rate 4%
$1/\sigma$	EIS	1	Standard
$1/\nu$	Frisch elasticity	0.5	Standard
ω	Intermediate inputs share	0.5	Itskhoki and Mukhin (2021a)
α	Openness	0.2	Imports-to-GDP ratio 40%
$\varkappa$	Capital share in value-added	0.43	Investment-to-GDP ratio 22%
δ_k	Capital depreciation rate	0.04	Penn World Table 10.0
ϕ_I	Investment adjustment cost	2.5	Christiano et al. (2005)
ϕ_π	Taylor coefficient	1.5	Standard
ρ_m	Monetary policy inertia	0.43	Smets and Wouters (2007)
η	Trade elasticity	1.5	Standard
$\bar{s}$	Foreign currency assets & liabilities	0.52	Bénétix et al. (2015)
ρ	Shock persistence	0.89	Itskhoki and Mukhin (2021a)
$\{\theta_{ij}^k\}$	Pricing regime	See text	LCP
Γ	Bond demand inverse elasticity	0.001	Itskhoki and Mukhin (2021a)
B. Estimated Parameters			
κ_p	Price Phillips curve slope	0.024	(0.006)
κ_w	Wage Phillips curve slope	0.010	(0.003)
h	Habit	0.819	(0.039)

- UIP deviation with the two shocks

$$\Omega(\{NFA_{kt}\}_k, \psi_{jt}, \{\zeta_{kt}\}_k) \equiv -\Gamma \left[\left(1 - \int \bar{s}_{ji} di \right) NFA_{jt} + \int \bar{s}_{ij} NFA_{it} di \right. \\ \left. + n^\psi \psi_{jt} + n^\zeta \left(- \int \bar{s}_{ji} di \zeta_{jt} + \int \bar{s}_{ij} \zeta_{it} di \right) \right]$$

where $n^\zeta = \left(\frac{1}{\Gamma^h} + \frac{1}{\Gamma^f} \right) \frac{\bar{\alpha}}{\beta}$

Back